MATH 1210 Worksheet 2 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

`http://home.cc.umanitoba.ca/~thomas/Courses`

1. Write the following complex numbers in Cartesian form.
   (a) $(2+4i)-(3-2i)$
   (b) $\overline{2+i}-(3+\overline{4i})$
   (c) $\frac{1}{i}$
   (d) $(1+2i)^2$
   (e) i^3-3i^2+2i+4
   (f) $(\overline{1+i})^2+\overline{(1+i)^2}$
   (g) $\frac{(1+i)^2(2-i)}{(3+2i)^2}$
   (h) $\overline{(\overline{4-i})^2}$
   (i) $\frac{1}{2i}\left(\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right)$
   (j) $\left(\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right)^4\left(\frac{1}{\sqrt{2}}-\frac{i}{\sqrt{2}}\right)^4$

Solution:

(a) $(2+4i)-(3-2i)=(2-3)+(4-(-2))i=-1+6i$

(b) $\overline{2+i}-(3+\overline{4i})=2-i-(3-4i)=(2-3)-(1-4)i=-1+3i$

(c) $\frac{1}{i}=\frac{-i}{i\cdot(-i)}=\frac{-i}{-i^2}=-i.$

(d) $(1+2i)^2=1^2+2\cdot 2i+(2i)^2=1+4i-4=-3+4i$

(e) $i^3-3i^2+2i+4=-i-3(-1)+2i+4=7+i$

(f) $(\overline{1+i})^2+\overline{(1+i)^2}=(1-i)^2+\overline{2i}=-2i-2i=-4i$

(g) $\frac{(1+i)^2(2-i)}{(3+2i)^2}=\frac{2i(2-i)}{5+12i}=\frac{2+4i}{5+12i}=\frac{(2+4i)(5-12i)}{(5+12i)(5-12i)}=\frac{58-4i}{169}=\frac{58}{169}-\frac{4}{169}i$

(h) $\overline{(\overline{4-i})^2}=\overline{(4+i)^2}=\overline{15+8i}=15-8i$

(i) $\frac{1}{2i}\left(\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right)=\frac{-i}{2}\left(\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right)=\frac{1}{2}\left(\frac{-i}{\sqrt{2}}+\frac{-i^2}{\sqrt{2}}\right)=\frac{1}{2\sqrt{2}}-\frac{i}{2\sqrt{2}}$

(j) $\left(\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right)^4\left(\frac{1}{\sqrt{2}}-\frac{i}{\sqrt{2}}\right)^4=\left|\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right|^8=\left|e^{i\pi/4}\right|^8=1$

2. Express the following complex numbers in polar form. Use the principal value for the argument.
   (a) $-1+i$
   (b) $-2i$
   (c) $\sqrt{3}-i$
   (d) $-\sqrt{3}-3i$
   (e) $-2\left[\cos(\pi/3)-\sin(\pi/3)i\right]$

Solution:

(a) The modulus of $-1+i$ is $\sqrt{2}$, the principal value for the argument of $-1+i$ is $3\pi/4$ and the polar form is $-1+i=\sqrt{2}\,(\cos(3\pi/4)+i\sin(3\pi/4))$.

(b) The modulus of $-2i$ is 2, the principal value for the argument of $-2i$ is $\pi/2$. Hence the polar form is $-2i = 2\,(\cos(-\pi/2)+i\sin(-\pi/2))$.

(c) The modulus of $\sqrt{3}-i$ is 2, the principal value for the argument of $\sqrt{3}-i$ is $-\pi/6$. Hence the polar form is $\sqrt{3}-i=2\,(\cos(-\pi/6)+i\sin(-\pi/6))$

(d) The modulus of $-\sqrt{3}-3i$ is $2\sqrt{3}$, the principal value for the argument of $-\sqrt{3}-3i$ is $-4\pi/3$. Hence the polar form is $-\sqrt{3}-3i=2\sqrt{3}2\,(\cos(-4\pi/3)+i\sin(-4\pi/3))$

(e) We have

$$\begin{aligned}
-2\,[\cos(\pi/3)-\sin(\pi/3)i] &= -2\cos(\pi/3)+2\sin(\pi/3)i\\
&= 2\cos(\pi-\pi/3)+2\sin(\pi-\pi/3)i\\
&= 2\cos(2\pi/3)+2\sin(2\pi/3)i\\
&= 2\,(\cos(2\pi/3)+2\sin(2\pi/3)i)
\end{aligned}$$

3. Find the Cartesian form of

$$z=\frac{i^{62}(\sqrt{2}+\sqrt{6}i)^8}{2^8(-\sqrt{6}-\sqrt{2}i)}.$$

Simplify as much as possible.

Solution: First let us find the polar form of $\sqrt{2}+\sqrt{6}i$ that we will need. The modulus is

$$|\sqrt{2}+\sqrt{6}i| = \sqrt{2+6}=\sqrt{8}=\sqrt{2}^3.$$

After dividing by the modulus we find that

$$\frac{\sqrt{2}+\sqrt{6}i}{|\sqrt{2}+\sqrt{6}i|}=\frac{\sqrt{2}+\sqrt{6}i}{\sqrt{2}^3}=\frac{1}{2}+\frac{\sqrt{3}}{2}i=e^{i\pi/3}.$$

Hence

$$\sqrt{2}+\sqrt{6}i=\sqrt{2}^3\,(\cos(\pi/3)+\sin(\pi/3)i)\,.$$

Using that $i^{62}=(i^2)^{31}=(-1)^{-31}=-1$ we find that

$$\begin{aligned}
z &= \frac{i^{62}(\sqrt{2}+\sqrt{6}i)^8}{2^8(-\sqrt{6}-\sqrt{2}i)}\\
&= \frac{(\sqrt{2}+\sqrt{6}i)^8}{2^8(\sqrt{6}+\sqrt{2}i)}\\
&= \frac{1}{i}\frac{(\sqrt{2}+\sqrt{6}i)^8}{2^8(\sqrt{2}-i\sqrt{6})}\\
&= \frac{1}{i}\frac{(\sqrt{2}+\sqrt{6}i)^8(\sqrt{2}+\sqrt{6}i)}{2^8(\sqrt{2}-i\sqrt{6})(\sqrt{2}+\sqrt{6}i)}\\
&= \frac{1}{i}\frac{(\sqrt{2}+\sqrt{6}i)^9}{2^8\cdot 8}\\
&= \frac{1}{i}\frac{(\sqrt{2}+\sqrt{6}i)^9}{2^{11}}. \qquad (1)
\end{aligned}$$

By De Moivre's formula we know that

$$\begin{aligned}
(\sqrt{2}+\sqrt{6}i)^9 &= \left[\sqrt{2}^3\left(\cos(\pi/3)+\sin(\pi/3)i\right)\right]^9 \\
&= \sqrt{2}^{27}\left(\cos(9\pi/3)+\sin(9\pi/3)i\right) \\
&= \sqrt{2}^{27}\left(\cos(\pi)+\sin(\pi)i\right) \\
&= -\sqrt{2}^{27}.
\end{aligned}$$

Substituting this in (1) we continue the computation

$$\begin{aligned}
\frac{1}{i}\frac{(\sqrt{2}+\sqrt{6}i)^9}{2^{11}} &= -\frac{1}{i}\frac{\sqrt{2}^{27}}{2^{11}} \\
&= -\frac{\sqrt{2}^5}{i} \\
&= -\frac{\sqrt{2}^5(-i)}{(-i)i} \\
&= i\sqrt{2}^5.
\end{aligned}$$

Hence the answer is $z = i\sqrt{2}^5$.

4. Find, in simplified form, the complex number z that satisfies

$$(1-i)^2(1+i)^2 + \overline{(4-3i)}z = 1+4i.$$

Solution: First, we have $(1-i)^2(1+i)^2 = |1+i|^4 = 4$. Hence the equation can be written as

$$\begin{aligned}
\overline{(4-3i)}z = -3+4i &\iff (4+3i)z = -3+4i \\
&\iff z = \frac{-3+4i}{4+3i} \\
&\iff z = i\frac{3i+4}{4+3i} \\
z = i.
\end{aligned}$$

Hence the solution is $z = i$.

5. Solve the following equations:
   (a) $x^2+2x+7=0$,
   (b) $x^4+4x^2+3=0$.

Solution:

(a) The discriminant is $2^2 - 4\cdot 7 = 4 - 28 = -24$. Hence the solutions are

$$x = \frac{-2\pm\sqrt{-24}}{2} = \frac{-2\pm\sqrt{-24}}{2} = -1\pm i\sqrt{6}.$$

(b) We set $y = x^2$. Then the equation becomes $y^2 + 4y + 3 = 0$ and is a quadratic equation that we can solve for y. The discriminant of this equation is $4^2 - 4 \cdot 3 = 16 - 12 = 4$. Hence the solutions of the quadratic equation are $y = -1$ and $y = -3$. Now the solutions x of the equation $x^4 + 4x^2 + 3 = 0$ are exactly the complex numbers such that $x^2 = y$ is a solution of the quadratic equation $y^2 + 4y + 3 = 0$, that is $x^2 = -1$ or $x^2 = -3$. It follows that the solutions of $x^4 + 4x^2 + 3 = 0$ are

$$x = \pm - i \quad \text{and} \quad x = \pm\sqrt{3}i.$$

6. Consider the quadratic equation $x^2 - tx + 1 = 0$ where t is a real number. For what values of t does the equation have two distinct complex conjugate solutions?

Solution: The discriminant of the equation is $t^2 - 4$. The equation has two distinct complex conjugate solutions precisely when the discriminant is negative, that is $t^2 - 4 < 0$. This means that $t^2 < 4$, hence $-2 < t < 2$.

7. Verify the following properties for complex numbers.
   (a) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$ for two complex numbers $z_1 = a + ib$ and $z_2 = x + iy$.
   (b) $\frac{z}{|z|}$ has modulus 1.
   (c) Let α be a nonzero real number. Then $\arg(\alpha z) = \arg(z) + 2k\pi$ if α is positive, and $\arg(\alpha z) = \pi + \arg(z) + 2k\pi$ if α is negative.

Solution:

(a) Let $z_1 = a + ib$ and $z_2 = x + iy$. Then

$$z_1 z_2 = (a + ib)(x + iy) = ax + ibx + iay + i^2 by = (ax - by) + i(bx + ay)$$

and its complex conjugate is

$$\overline{z_1 z_2} = (ax - by) - i(bx + ay).$$

On the other hand we have

$$\bar{z}_1 \bar{z}_2 = (a - ib)(x - iy) = ax - ibx - iay + i^2 by = (ax - by) - i(bx + ay),$$

which is the same as $\overline{z_1 z_2}$.

(b) We have seen that $|z_1/z_2| = |z_1|/|z_2|$. Hence

$$\left|\frac{z}{|z|}\right| = \frac{|z|}{||z||} = \frac{|z|}{|z|} = 1.$$

The modulus of $|z|$ is $|z|$ because it is a real number.

(c) When we multiply a complex number z by a positive real number α, the segment joining 0 and αz is in the same direction than the segment joining 0 and z. Hence the angles are the same, up to multiples of 2π. When α is negative, then the segment joining 0 and αz is in the opposite direction to the segment joining 0 and z. Hence $\arg(\alpha z)$ and $\arg(z)$ differ by π, and a multiple of two.
Alternatively, you can use the formula we saw in class:

$$\arg(\alpha z_2) = \arg(\alpha) + \arg(z_2) + 2k\pi.$$

Then $\arg(\alpha) = 0$ if $\alpha > 0$, and $\arg(\alpha) = \pi$ if $\alpha < 0$.